

Khawaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Fall Semester 2018

Program: B.Sc. in CSE (Batch: 6th) 1st Year 2nd Semester

Course Title: Physics – II

Course Code: PHY 1201

Total Time: 2 hours

Full Marks: 40

Group-B

[Answer any 5 (Five) questions including No.1]

5 × 6 = 30

1. Read the stem given below and answer the following questions:

A radioactive substance is ^{222}Rn , whose half life is 3.82 days. But it is obvious that the radioactivity never ends.

- | | |
|--|---|
| a) Define half-life. | 1 |
| b) Find the value of activity of 1gm of above mentioned substance. | 3 |
| c) Justify the last statement of above stem. | 2 |
| 2. a) Show by the theory of relativity that the velocity of a body cannot be more than the velocity of light. 2 | |
| b) What is the length of a meter stick moving parallel to its length when its mass is 3/2 of its rest mass? | 4 |
| 3. a) Write down the inverse Lorentz transformation equations. 2 | |
| b) An astronaut at the age of 25 years went to the space by a space-shuttle moving with $1.8 \times 10^8 \text{ms}^{-1}$ velocity. According to earth he spent 30 years in space. What is his age now? | 4 |
| 4. a) Define Work function & Stopping potential. 2 | |
| b) Draw the incident light frequency vs energy curve of photoelectric effect. | 1 |
| c) The wavelength of light falling on the surface of a metal of work function 2.3 eV is $4300 \text{\AA}$ with what velocity the electron will be emitted? | 3 |
| 5. a) What is nucleon? 1 | |
| b) Write a short note on "beta decay is a nuclear phenomenon" | 2 |
| c) Calculate the radius of the third orbit of hydrogen atom. Also calculate the energy of electrons in that orbit. 3 | |
| (Given $h = 6.62 \times 10^{-34} \text{JS}$, $m = 9.1 \times 10^{-31} \text{kg}$, $e = 1.6 \times 10^{-19} \text{C}$) | |
| 6. a) Define mass defect. 1 | |
| b) Draw a block diagram of a nuclear power plant. 2 | |
| c) Find the Binding Energy of ^4_2He nucleus. 3 | |
| [$m_p = 1.00758 \text{ a.m.u}$; $m_n = 1.00894 \text{ a.m.u}$; $M_{\text{He}} = 4.00389 \text{ a.m.u}$] | |

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$$0.5 \times 20 = 10$$

Student Name:	ID No.:
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- What is the radius of the first Bohr orbit for H atoms?
 - $0.13A^{\circ}$
 - $0.23A^{\circ}$
 - $0.33A^{\circ}$
 - $0.43A^{\circ}$
 - $0.53A^{\circ}$
- When an electron jumps from the outer orbits to the first orbit, this corresponds to
 - Lyman series
 - Balmer series
 - Paschen series
 - Bracket series
 - Pfund series
- Mass of proton is –
 - $1.67 \times 10^{-27}kg$
 - $1.67 \times 10^{-34}kg$
 - $1.67 \times 10^{-19}kg$
 - $9.1 \times 10^{-27}kg$
 - $9.1 \times 10^{-31}kg$
- is electrically neutral.
 - Electron
 - Proton
 - Neutron
 - Alpha particle
 - Beta particle
- How many electrons in Alfa particle?
 - 4
 - 3
 - 2
 - 0
 - 1
- The radius of the ${}_{13}^{27}Al$ nucleus is –
 - 2.6fm
 - 3.6fm
 - 4.6fm
 - 5.6fm
 - 6.6fm

7. How much energy releases in every nuclear fusion?
 - a) 200MeV
 - b) 198MeV
 - c) 36.6MeV
 - d) 17.6MeV
 - e) 13.6MeV

8. The volume of a nucleus is to the number of nucleons contained in it.
 - a) proportional
 - b) inversely proportional
 - c) equal
 - d) independent
 - e) less than

9. What is the Compton wavelength?
 - a) $0.12A^0$
 - b) $0.012A^0$
 - c) $0.24A^0$
 - d) $0.024A^0$
 - e) $0.044A^0$

10. Acceleration of inertial frame of reference is –
 - a) zero.
 - b) infinity.
 - c) positive.
 - d) negative.
 - e) undefined.

11. The nuclear reaction, ${}^{35}_{17}\text{Cl} + {}^1_1\text{H} \rightarrow {}^{32}_{16}\text{S} + ?$
 - a) ${}^1_1\text{H}$
 - b) ${}^4_2\text{He}$
 - c) ${}^2_1\text{H}$
 - d) ${}^1_0\text{n}$
 - e) ${}^3_1\text{H}$

12. The ratio of radioactive decay per second to the total number of atoms is called the –
 - a) activity.
 - b) half life.
 - c) decay constant.
 - d) mean life.
 - e) decay curve.

13. Relation between total energy and momentum is –
 - a) $E = mc^2$
 - b) $E = mc$
 - c) $E = pc^2$
 - d) $E = pc$
 - e) $E = mp$

14. The kinetic energy of the ejected electron depends on the.....of incident photon.
 - a) photo electron
 - b) photo current
 - c) direction
 - d) intensity
 - e) frequency

15. $1\text{Curry} = ?$
 - a) $1Bq$
 - b) 10^6Bq
 - c) $10^{10}Bq$
 - d) 3.7×10^6Bq
 - e) $3.7 \times 10^{10}Bq$

16. When the velocity v of the body is negligibly small as compared to light velocity c , then which one will be correct relationship between moving mass (m) & rest mass (m_0)?
 - a) $m = m_0$
 - b) $m > m_0$
 - c) $m < m_0$
 - d) $m \gg m_0$
 - e) $m \ll m_0$
17. After decaying Alpha particle from ${}^{238}_{92}U$, which one is produced?
 - a) ${}^{233}_{89}Pd$
 - b) ${}^{234}_{90}Th$
 - c) ${}^{235}_{92}U$
 - d) ${}^{231}_{82}Rd$
 - e) ${}^{242}_{90}Th$
18. The disintegration constant of a radioactive element is 0.00231 per day. What is the average life of it?
 - a) 931.9 days
 - b) 707.9 days
 - c) 586.9 days
 - d) 432.9 days
 - e) 332.9 days
19. The slope of the energy and frequency curve for photo-electric effect is equal to the –
 - a) photo current
 - b) photo electron
 - c) plank constant
 - d) work function
 - e) stopping potential
20. Nuclei having same mass number but different atomic number/proton number are called –
 - a) isotope.
 - b) isobar.
 - c) isotone.
 - d) isomer.
 - e) mirror nucleus.

Invigilator Signature

Examiner Signature